

Q1. What exactly is the problem?

The problem is that vessel charter rates fluctuate unpredictably, so officials may book ships when prices are high. We need to forecast future rates and identify the right time to charter, reducing unnecessary freight costs.

Q2. Who exactly is facing this problem?

The primary users are the Ministry of Steel and government procurement teams.

They need to charter large vessels to import coal and iron ore.

Charter rates fluctuate significantly with market conditions.

Booking at the wrong time can result in higher transportation costs.

They need a reliable way to predict future freight-rate movements.

Our system helps them choose when to charter at a more favourable rate.

Q4. How is this problem currently being solved?

Currently, vessel chartering is mainly handled through daily market monitoring, broker quotations, freight indices, and expert judgment. Officials compare available spot rates for different vessels and routes based on immediate cargo requirements. Historical freight data and market trends may be used for basic estimation, while port restrictions and vessel suitability are checked separately. Factors such as fuel prices, congestion, and market conditions are considered manually from different sources. As a result, decisions are mostly reactive, with limited ability to predict the best booking time or optimize multiple-voyage contracts.

Q5. What are the limitations of existing solutions?

Existing solutions are mainly reactive and depend on current freight rates and market trends. They lack accurate route- and vessel-specific forecasting. Important factors like fuel

prices, port congestion, exchange rates, weather, and geopolitical risks are not fully integrated. Port infrastructure and vessel compatibility are often handled separately. They provide limited support for reducing idle time, demurrage, and overall logistics costs. Most importantly, they lack explainable, actionable recommendations for optimal chartering decisions.

Q6. Why does it need a new solution?

A new solution is needed to shift vessel chartering from reactive decisions to predictive and data-driven planning. Existing methods cannot accurately forecast freight rates while considering fuel prices, port congestion, vessel availability, exchange rates, and market risks together. The proposed solution can recommend the right vessel, optimal booking time, and feasible routes while reducing idle time, demurrage, and overall transportation costs. It will also provide explainable insights to help officials make faster, safer, and more cost-effective chartering decisions.

Q7. What is our solution and does it fulfill the

PS requirements?

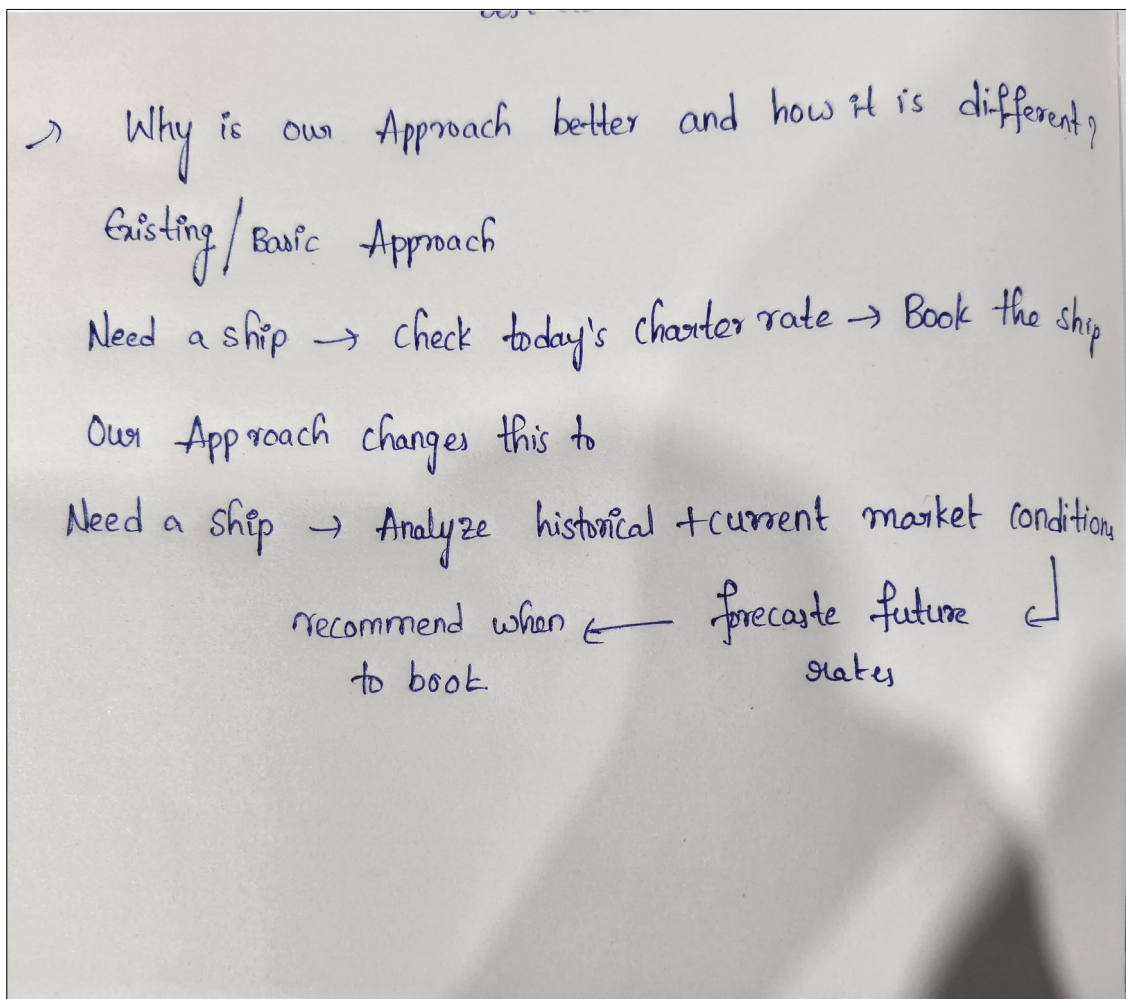
Our solution is an Intelligent Freight Forecasting and Vessel Chartering Decision-Support System that uses historical and real-time data to forecast freight rates and recommend the best time to charter a vessel. It integrates freight rates, bunker fuel prices, vessel supply, exchange rates, port congestion, weather, commodity demand, and port infrastructure constraints. Based on cargo, origin, destination, and vessel requirements, it recommends the most suitable vessel type, optimal market-entry timing, and strategies to reduce idle time and demurrage. The system also provides risk alerts, cost comparisons, forecast confidence, and explainable AI insights through an interactive dashboard. Yes, it fulfills the SIH26006 requirements by directly addressing freight forecasting, optimal charter timing, vessel selection, port constraints, idle-time management, risk mitigation, and the shift from spot contracts toward short- and medium-term multiple-voyage contracts.

Q8.

→ What is the innovation that are we doing in our solution
Historical freight rates + market factors → forecast
↓
← Explain the reason
Recommended best action.

Our innovation is that we are not only forecasting vessel freight/charter rates, but converting that forecast into an actionable procurement decision for Indian bulk-material imports.

Q9.



Our approach is better because it is predictive rather than reactive. Instead of checking the freight rate only when a ship is needed, we use historical and current market factors to forecast how the rate may change. Then we convert that forecast into a practical booking recommendation, while considering the cargo and port requirements. So we're not just predicting a number—we're supporting the actual chartering decision.

Q10. How can we demonstrate the measurable impact of our

solution?

We can measure the impact by comparing our system with the current manual/spot-chartering approach using historical data and simulated chartering decisions. Key metrics include freight cost reduction (%), forecasting accuracy (MAE/RMSE), reduction in vessel idle time and demurrage, improvement in vessel utilization, and savings from optimal booking timing. We can also measure how accurately the system selects feasible vessel types and predicts port congestion and market risks. A back-testing module can show how much cost could have been saved if our recommendations were followed in previous years. These results can demonstrate improved cost efficiency, planning accuracy, and reliability for the Ministry of Steel.

Q11. What are the required datas and is the datas and sources are available or not?

Our solution requires historical freight rates, bunker fuel prices, vessel supply data, USD/INR exchange rates, port congestion, AIS tracking, commodity prices, weather,

geopolitical events, and port infrastructure data. Most of these datasets are publicly available or accessible through APIs and maritime data providers. However, real-time port congestion, detailed charter-party rates, and some AIS data may have limited or paid access. Therefore, the prototype can be developed using publicly available historical datasets and simulated/representative operational data where necessary, while the full production system can integrate authorized real-time data sources.

Required Data	Example Source
Historical freight rates / BDI	Baltic Exchange
Bunker fuel prices (VLSFO/MGO)	Ship & Bunker, Bunker Index providers
Vessel fleet & orderbook	Clarksons Research , UNCTAD
USD/INR exchange rate	Reserve Bank of India financial APIs
Port congestion / waiting time	Port authorities, AIS-based providers
AIS vessel tracking	MarineTraffic , VesselFinder
Coal & iron ore prices	World Bank , commodity exchanges
Weather / cyclone data	India Meteorological Department , NOAA
Port infrastructure	Indian port authorities, Ministry of Ports, Shipping and Waterways
Geopolitical events	GDELT, news/event databases
Historical charter/fixture rates	Shipping brokers / maritime databases

